CHAPTER

20

Chebyshev Filters

Chebyshev filters are used to separate one band of frequencies from another. Although they cannot match the performance of the windowed-sinc filter, they are more than adequate for many applications. The primary attribute of Chebyshev filters is their speed, typically more than an order of magnitude faster than the windowed-sinc. This is because they are carried out by *recursion* rather than *convolution.* The design of these filters is based on a mathematical technique called the *z-transform,* discussed in Chapter 33. This chapter presents the information needed to *use* Chebyshev filters without wading through a mire of advanced mathematics.

The Chebyshev and Butterworth Responses

The Chebyshev response is a mathematical strategy for achieving a faster *roll-off* by allowing *ripple* in the frequency response. Analog and digital filters that use this approach are called *Chebyshev filters*. For instance, analog Chebyshev filters were used in Chapter 3 for analog-to-digital and digital-to-analog conversion. These filters are named from their use of the *Chebyshev polynomials,* developed by the Russian mathematician Pafnuti Chebyshev (1821-1894). This name has been translated from Russian and appears in the literature with different spellings, such as: Chebychev, Tschebyscheff, Tchebysheff and Tchebichef.

Figure 20-1 shows the frequency response of low-pass Chebyshev filters with passband ripples of: 0%, 0.5% and 20%. As the ripple increases (bad), the roll-off becomes sharper (good). The Chebyshev response is an optimal trade-off between these two parameters. When the ripple is set to 0%, the filter is called a **maximally flat** or **Butterworth filter** (after S. Butterworth, a British engineer who described this response in 1930). A ripple of 0.5% is a often good choice for digital filters. This matches the typical precision and accuracy of the analog electronics that the signal has passed through.

The Chebyshev filters discussed in this chapter are called **type 1** filters, meaning that the ripple is only allowed in the *passband.* In comparison,